

Technical Report

Building a Production-Ready Credit Risk Classification System

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1. Executive Summary

In an effort to provide real-time credit risk classification, this report explains the entire process from conceptualization to production candidate implementation of a binary model to classify whether or not an applicant will default on a loan. The entire project followed a highly integrated methodology consisting of three distinct phases:

- 1) Data exploration and preparation (preprocessing).
- 2) Development of the credit risk classifier (model development), and determination of Classifier hyper-parameters (hyper parameter tuning).
- 3) Final evaluation and preparation for systems deployment. To train and evaluate the model(s), a synthetic dataset consisting of 1,000 applicants was created with domain realistic demographics, account health information, and characteristics of the loan that was requested.

During this process, four different algorithms were trained and compared using Stratified K-Fold Cross-Validation: Logistic Regression, Decision Trees, Random Forest, and XGBoost. The XGBoost algorithm outperformed the other algorithms with a classification accuracy of 87.5%, macro F1 score of 0.87, and an AUC-ROC score of 0.9408 across the validation dataset. All experiments conducted throughout the classification model development process were logged in MLflow to prove that the results could be reproduced.

Key Outcome

XGBoost selected as the production model — Test Accuracy 87.5% | AUC-ROC 0.9408 | Default Recall 82.3% — with a recommended deployment via FastAPI + Docker + Kubernetes.

2. Dataset Overview

2.1 Data Generation

The dataset was generated synthetically using NumPy with a fixed random seed (42), thus ensuring total reproducibility. The risk labels were assigned based on a pre-determined rule that mimics actual credit evaluation: risk points are attributed to the applicant based on their age (too young or too old), the size of the loan being requested (too high), the length of the loan (too long), the account balance of the applicant (too low), and whether or not the applicant is unskilled.

Property	Detail	Notes
Sample size	1,000 rows	80% train / 20% test, stratified
Features	9 input features	4 numeric, 5 categorical (incl. target)
Target classes	Good (0) — 604 / Bad (1) — 396	Mild class imbalance (~60/40)
Missing values	Saving acc.: 85 Checking acc.: 95	Treated as informative, not imputed

2.2 Feature Description

The raw feature set covers four dimensions of an applicant's financial profile:

- Demographics: Age, Sex
- Employment: Job (0 = unskilled/non-resident through 3 = highly skilled)
- Housing status: own / rent / free
- Account health: Saving accounts and Checking account (ordinal), both with meaningful missing-value rates
- Loan characteristics: Credit amount (₹298 – ₹19,972), Duration (4 – 72 months), and Purpose (8 categories)

3. Exploratory Data Analysis

3.1 Univariate Distributions

Age follows a broad, roughly uniform distribution with a slight peak around 50–60, indicating that middle-aged applicants dominate the applicant pool. Credit amount is right-skewed — the majority of loans cluster in the moderate range while a tail of high-value loans pulls the mean (₹9,944) well above the median (₹9,814). Loan duration clusters near 36 months, suggesting medium-term borrowing is the norm.

3.2 Bivariate Risk Analysis

The group-level analysis surfaced three strong predictive signals that directly shaped feature engineering decisions:

- Credit amount: Bad-risk applicants borrow ₹12,080 on average versus ₹8,543 for good-risk — a 41% gap. Larger loans represent bigger financial commitments and therefore higher failure probability.
- Loan duration: Bad-risk applicants carry a mean duration of 42.9 months versus 32.8 months for good-risk — a 13% longer repayment window during which circumstances can deteriorate.
- Savings disclosure: Applicants who did not disclose savings defaulted at 58.8%, nearly 21 percentage points above those who did (37.8%). The silence itself is a predictive signal.

Insight

Age showed virtually no separation between risk classes (mean bad: 46.9, good: 45.8), and loan purpose explained little beyond its frequency distribution. Both were treated accordingly in feature engineering.

3.3 Correlation Structure

A Pearson correlation heatmap of the four numerical features revealed no strong linear relationships — the highest pair-wise correlation was between Job and Duration at a statistically negligible level. This

confirms that each numeric feature contributes independent information and that dropping any on grounds of collinearity was unwarranted.

3.4 Missing Value Analysis

Of 1,000 rows, 85 are missing Saving accounts and 95 are missing Checking account, with 6 rows missing both (180 flags total). The missingness mechanism was investigated and confirmed to be informative rather than random: applicants who withheld savings information defaulted at a markedly elevated rate. Imputing these values would destroy this signal; they were therefore retained as an explicit categorical level.

4. Feature Engineering

Five new features were constructed before preprocessing to encode domain knowledge and improve model signal quality. A High_Risk_Purpose flag was considered but discarded after EDA showed that purpose-specific default rates were proportionate to their frequency — the variable carried no independent predictive power.

Feature	Description	Rationale
Monthly_Burden	Credit Amount / Duration	Captures combined repayment pressure in a single scalar; both components individually correlated with risk.
Age_Group	Age binned into 4 ordinal groups	EDA revealed a non-linear (U-shaped) age-risk relationship; raw age was therefore less informative than bins.
Account_stability	Ordinal sum of saving + checking balances (0-7)	Consolidates two correlated account-health signals while penalising missing disclosures at zero.
Saving_missing	Binary flag for undisclosed savings	Missingness is predictive: undisclosed savers default at 58.8% vs 37.8% for disclosed ones.
Checking_missing	Binary flag for undisclosed checking	Weaker signal (42% vs 39%) but retained for completeness.

5. Preprocessing Pipeline

All transformations were applied inside a scikit-learn ColumnTransformer to prevent data leakage — the transformer was fit only on the training set and then applied identically to the test set. This is a non-negotiable requirement for honest model evaluation.

Transformation	Columns	Justification
Ordinal Encoding	Saving accounts, Checking account	Natural ordered hierarchy exists (little < moderate < quite rich < rich); missing values mapped to -1.
One-Hot Encoding	Sex, Housing, Purpose	Nominal categories with no intrinsic order; first category dropped to avoid multicollinearity.
Standard Scaling	All numeric features	Age and Credit Amount operate on vastly different scales; scaling ensures no feature dominates gradient-based optimisers.

Raw Age was dropped in favour of Age_Group because EDA showed a U-shaped risk pattern (younger and older applicants both elevated) that linear ordinal scaling would misrepresent. The transformed training matrix contained 18 features after one-hot expansion. Data was split 80/20 with stratification to preserve the 60/40 class ratio in both partitions.

6. Model Development

6.1 Baseline Models

All four classifiers were first evaluated without any hyperparameter tuning to establish honest baselines. Logistic Regression and Decision Tree provided the first useful signal; the ensemble methods immediately outperformed them in accuracy and recall. The Decision Tree, as expected, memorised the training set perfectly (100% training accuracy) while achieving only 80.5% on the test set — a classic sign of overfitting.

6.2 Hyperparameter Tuning

GridSearchCV was applied to each model with Stratified K-Fold (k=5) cross-validation. Stratification was chosen deliberately: because this is a binary classification task with mild class imbalance, standard K-Fold can produce folds where one class is over- or under-represented, inflating CV scores and producing unreliable parameter selection. Stratified K-Fold ensures that each fold preserves the original 60/40 class distribution.

Why Stratified K-Fold?

In each fold, the Good-to-Bad ratio mirrors the full dataset (60:40). This gives a more reliable estimate of generalisation performance and eliminates the bias that random splits can introduce in imbalanced classification tasks.

Logistic Regression

Tuned over regularisation strength (C) and solver type. The model is intentionally simple — it acts as an interpretable baseline and a sanity check for the ensemble methods.

Decision Tree

Max depth and minimum samples per split were tuned to constrain overfitting. Even after tuning, the training-test gap of 0.13 persisted, confirming that single trees are structurally prone to high variance on this dataset.

Random Forest

Number of estimators, max depth, and minimum samples per leaf were searched. The ensemble mechanism substantially reduced variance relative to the single tree — training accuracy remained high at 0.99 but the generalisation gap narrowed to 0.13.

XGBoost

Learning rate, max depth, and number of estimators were tuned via stratified grid search. The best configuration achieved a training accuracy of 0.9487 and test accuracy of 0.875 — the most balanced result across all models, with a generalisation gap of approximately 0.07.

7. Model Evaluation Results

7.1 Performance Summary

All metrics are reported on the held-out test set (n=200) after stratified cross-validation tuning:

Model	Train Acc	Test Acc	F1 (avg)	AUC-ROC	Gen. Gap
Logistic Regression	0.84	0.775	0.83	~0.84	0.01
Decision Tree	0.94	0.820	0.80	~0.82	0.13
Random Forest	0.99	0.850	0.86	~0.90	0.13
XGBoost (Tuned)	0.9487	0.875	0.87	0.9408	0.07

7.2 Confusion Matrix Analysis

Confusion matrices were plotted for all four models. XGBoost correctly identified 124 out of 124 non-defaulters flagged (True Negatives) and 76 of 200 test samples' defaulters. XGBoost and Random Forest displayed the best balance between catching true defaulters and minimising false alarms. The Decision Tree, despite its tuning, produced a slightly higher misclassification rate. From a business perspective, False Negatives — missed defaulters — are substantially more costly than False Positives and were tracked separately throughout the evaluation.

7.3 ROC-AUC Analysis

XGBoost achieved an AUC of 0.9408, the highest of all models. Random Forest followed at approximately 0.90, with Logistic Regression and Decision Tree trailing behind. An AUC above 0.93 means the model correctly ranks a random defaulter above a random non-defaulter more than 94% of the time — a strong result for a production credit scoring system.

7.4 Precision-Recall Analysis

The Precision-Recall curve for XGBoost was evaluated with an operating point set at recall ≥ 0.75 for the default class. The model maintained strong precision at this operating point, confirming that its high recall does not come at the cost of an unacceptable false-alarm rate. The Average Precision score further validated robust detection capability across all classification thresholds.

7.5 Feature Importance

XGBoost feature importances highlighted four dominant predictors: Account_stability, Job category, Saving accounts (encoded), and Credit amount. This aligns strongly with domain intuition — financial stability and borrowing behaviour are the primary drivers of creditworthiness. The engineered Account_stability feature, which consolidated saving and checking account health, ranked highest, validating the feature construction decision.

Random Forest distributed importance more evenly across features (reflecting its averaging mechanism), with Credit amount, Job, and Duration as top contributors. The Decision Tree concentrated importance heavily on a small subset of splits — a hallmark of its structural tendency to overfit individual data patterns.

8. Business Interpretation

8.1 Why Recall Is the Primary Metric

In credit risk, the cost asymmetry between error types is fundamental. A False Negative — predicting a future defaulter as safe and approving the loan — results in the loss of the full principal plus collection costs. A False Positive — declining a creditworthy applicant — costs only the opportunity to earn interest income on that loan.

Research and practitioner experience suggest that the cost of a missed default is 5 to 20 times larger than the cost of a wrongly rejected applicant. A 10:1 cost ratio means a model optimised purely for accuracy can appear impressive while being economically destructive if it systematically misses defaulters. This is why default-class recall was used as the primary model selection criterion, with AUC-ROC as the secondary discriminator.

Cost Perspective

Assuming an average credit amount of ₹10,000 and a 10% default rate, missing just 5 additional defaulters per 1,000 loans translates to ₹50,000 in incremental losses. A 3–5% improvement in recall can materially reduce expected portfolio risk.

8.2 Threshold Calibration

The default classification threshold of 0.5 is a starting point, not a business decision. Regulators and credit officers can shift the operating threshold along the Precision-Recall curve to reflect their specific risk appetite. A more conservative institution may set the threshold lower to maximise recall at the cost of some precision; a growth-oriented lender may accept more False Negatives in exchange for approving more borderline applicants.

9. Model Selection

XGBoost was selected as the production model on the basis of four converging criteria:

- Highest default-class recall (82.3%) — the metric that most directly maps to financial loss avoidance.
- Highest AUC-ROC (0.9408) — confirming genuine discriminative power rather than threshold gaming.
- Smallest generalisation gap (~0.07) among ensemble models — indicating the model is not memorising training idiosyncrasies.
- Native feature importance output — enabling regulatory-grade explanations when an applicant challenges a decline decision.

Logistic Regression remains the most interpretable model line by line, but its inability to capture non-linear relationships limits recall. Decision Tree is intuitive but overfits. Random Forest is a strong runner-up — in a latency-constrained environment where XGBoost's marginal gain does not justify deployment complexity, Random Forest would be the natural fallback.

Production Recommendation
XGBoost (Tuned) — Test Acc: 87.5% AUC: 0.9408 Default Recall: 82.3% Deployment: FastAPI + Docker + Kubernetes

10. Deployment Plan

10.1 Infrastructure

The selected model would be wrapped in a FastAPI service for its asynchronous performance and auto-generated OpenAPI documentation. The service would be containerised with Docker and deployed on Kubernetes to enable horizontal scaling during peak origination periods — typically month-end and salary-credit windows. Model artefacts, including preprocessing pipelines and evaluation snapshots, would be versioned in MLflow so that every production inference is traceable back to a specific training run.

10.2 Experiment Tracking (MLflow)

All four model runs were logged to MLflow across the development phase, capturing hyperparameters, per-class metrics, confusion matrix artefacts, and serialised model objects. This provides a complete audit trail and enables the team to roll back to a previous model version within minutes if production drift is detected.

10.3 Monitoring & Drift Detection

Post-deployment, the system should monitor input feature distributions (PSI — Population Stability Index) and prediction score distributions on a rolling basis. Significant drift in account balance features or loan amount distributions is a leading indicator that the model's learned boundaries may no longer reflect the current applicant pool. A scheduled retraining cadence (quarterly, or triggered by PSI breach) should be established.

10.4 Explainability

For regulated lending environments, SHAP (SHapley Additive exPlanations) values should be integrated alongside the XGBoost predictions. SHAP provides per-prediction feature attributions that can translate directly into human-readable decline reasons — a requirement under consumer credit regulations in most jurisdictions.

11. Conclusions

This project demonstrates that a rigorous machine learning workflow — grounded in domain-aware feature engineering, leak-proof preprocessing, and statistically sound validation — can produce a credit risk classifier that performs competitively with industry benchmarks on a moderately complex dataset.

The most impactful decisions were not algorithmic. Treating missing savings disclosures as informative rather than imputing them, engineering `Account_stability` as a single composite score, and using Stratified K-Fold to produce honest cross-validation estimates all contributed more to final model quality than simply switching between algorithms.

XGBoost's AUC of 0.9408 and default recall of 82.3% represent a meaningful starting point for production deployment, with clear pathways to further improvement through threshold calibration, SHAP-based explainability, and ongoing drift monitoring.

References & Acknowledgements

Standard implementation patterns for `GridSearchCV`, `StratifiedKFold`, and `Pipeline` usage were drawn from the official scikit-learn documentation. MLflow logging conventions follow the official MLflow Python API reference. All experimental design, model comparison, feature engineering decisions, and performance analysis were conducted independently by the author.